BugC2

SKU: K033-C





















Description

BugC2 is a programmable robot base compatible with the M5StickC series controllers. This base integrates a four-channel L9110S motor driver and DC geared motors, with the STM32F030F4 as its core control chip. It also features two programmable RGB LEDs, an infrared encoder, and a dedicated battery holder. Through its top slot, BugC2 can be tightly connected to the M5StickC controller, receiving precise control commands via the I2C protocol (address 0x38). In addition to providing basic mobility, BugC2 also includes advanced features such as a Type-C charging interface, battery reverse connection protection, and voltage detection, ensuring safety and convenience. Its built-in 700mAh 16340 rechargeable lithium battery ensures long-lasting operation, making it a versatile and efficient programmable robot platform suitable for various educational and innovative applications.

Note

Foot Spring Protective Sleeve Note

The transparent plastic film on the outside of the BugC2 foot springs is designed to prevent the springs from falling off. Do not remove it, otherwise the springs may detach and affect normal device operation.

Features

- Programmable robot
- Remote control
- Four-channel motor driver (L9110S)
- 2x RGB LED
- Infrared remote decoding
- Battery voltage detection
- Onboard Type-C battery charging
- Battery reverse connection protection

Includes

- 1 x StickC Plus2
- 1 x BugC2 base
- 12 x Silicone foot pads

Applications

- Remote motor control

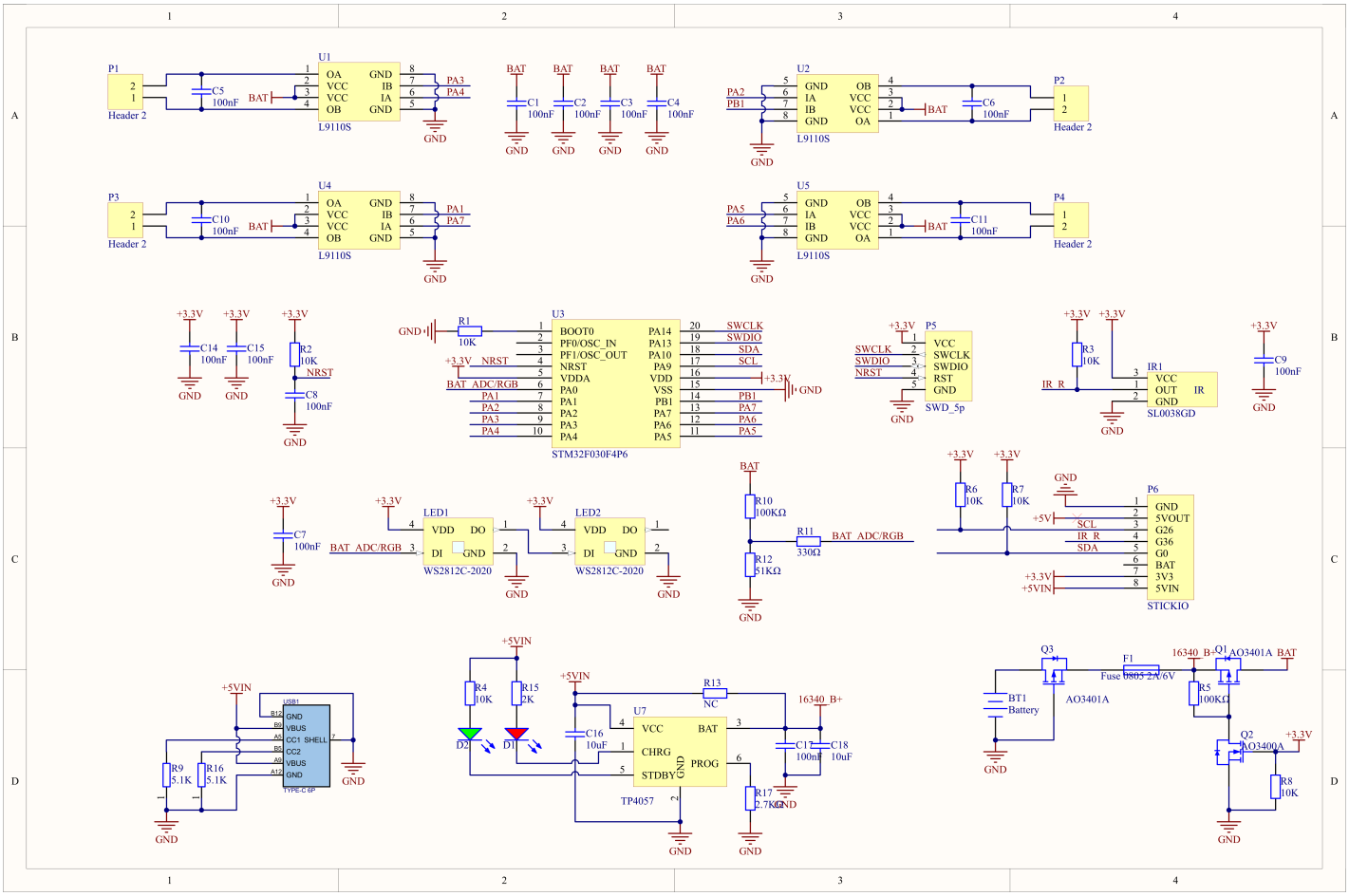
- Robot control
- Smart toys

Specifications

Specification	Parameter
MCU	STM32F030F4P6
Motor driver	L9110S
Infrared receiver	SL0038GD
RGB	WS2812C-2020
Battery capacity	700mAh (Model: 16340)
Standby current	Avg: DC4.2V@20.8uA
Operating current	Avg: DC4.2V@285.1mA
IR detection range (StickC Plus2)	∠180° infrared emission range (straight distance): 560CM ∠90° infrared emission range: 400CM
Operating temperature	0~40°C
Product Size	53.5 x 53.5 x 51.5mm
Product weight	74.5g
Package Size	71.7 x 70 x 58mm
Gross Weight	87.9g

Schematics

- BugC2 Schematics PDF



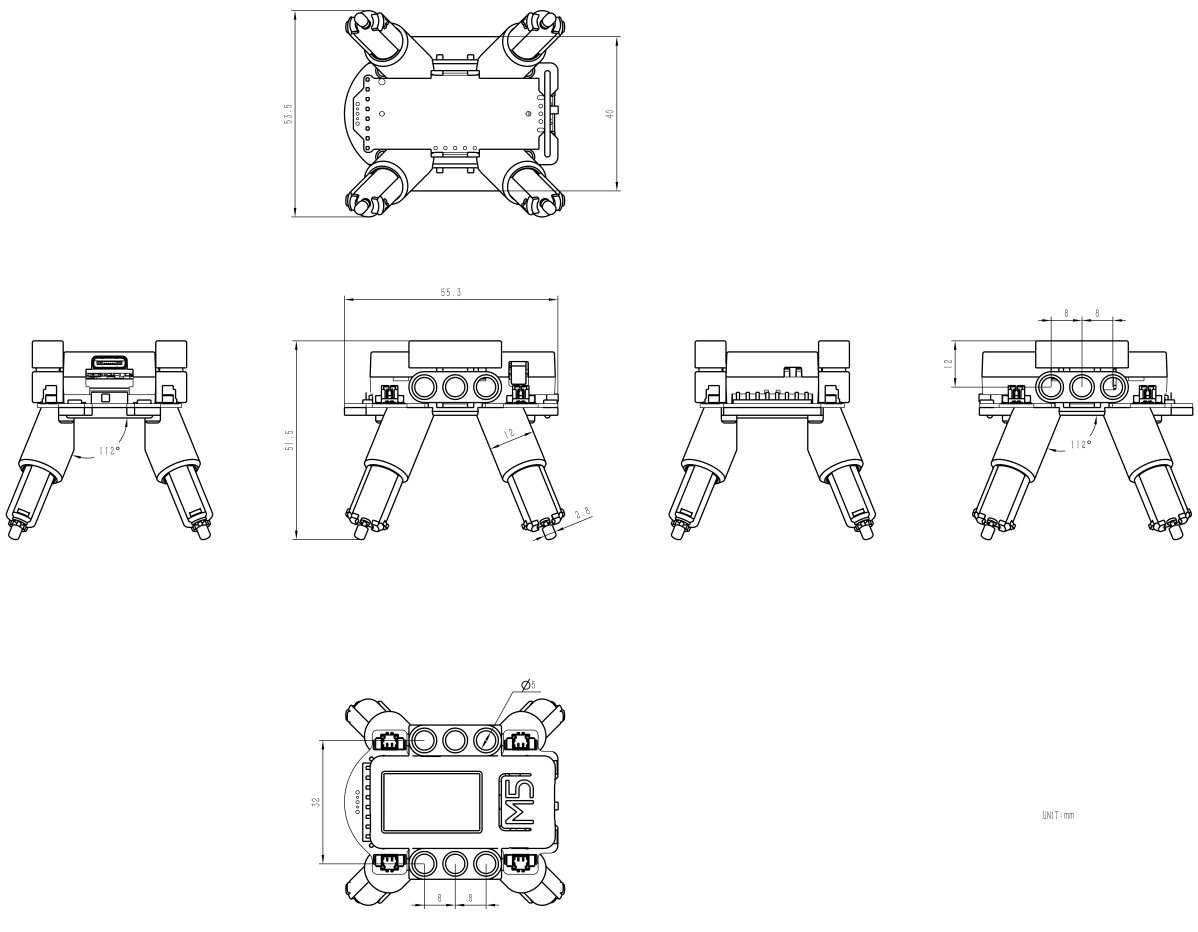
PinMap

STM32F030F4P6	PA1	PA2	PA3	PA4	PA1	PB1	PA5	PA6	PA7	PA0
Motor1 (left front)			1B	1A						
Motor2 (Left Rear)	1B								1A	
Motor3 (Right Rear)							1A	1B		
Motor4 (Right front)		1A				1B				
BAT_Detect (WS2812)										BAT_ADC/RGB

STICKC PLUS2 (ESP32-PICO-V3-02)	G36
IR (SL0038GD)	IR_R

Model Size

- BugC2 Model Size PDF



Datasheets

- L9110S

Softwares

Arduino

- BugC2 Arduino Library

UiFlow1

- Hat BugC2 Test Example

Internal Firmware

- BugC2 Internal Firmware

Protocol

M5Stack BugC I2C Protocol																	V1 (FW Version) 2022/11/29
REG MAP(Addr:0x38)		0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	note
MOTOR SPEED	0x00 W/R	front_left_speed	front_right_speed	rear_left_speed	rear_right_speed													speed: 0~-100; 0~100
RGB	0x10 W	RGB select	RGB_R	RGB_G	RGB_B													RGB select: 0: FRONT_LEFT RGB 1: FRONT_RIGHT RGB
ADC 8Bits	0x20 R	ADC 8Bits																0~255
ADC 12Bits	0x30 R	ADC 12Bits-L	ADC 12Bits-H															0~4095
Firmware Version	0xF0 R															Version		Version: firmware version number
I2C Address	0xF0 R/W																Address	Address: 1~127

Easyloader

Easyloader	Download Link	Notes
BugC2 Test Example Easyloader	download	/

Video

BugC2.mp4